

Chun-Yi Chang

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Education

Michigan State University, Ph.D. in Statistics, GPA: 3.9/4.0	Sep 2022 - Aug 2027
• Advisor: Dr. Chih-Li Sung • Coursework: Statistical Learning and Data Mining, Bayesian Inference	
Michigan State University, M.S. in Applied Statistics, GPA: 3.9/4.0	Sep 2020 - May 2022
National Taipei University, B.B.A in Statistics, GPA: 3.4/4.0	Sep 2016 - Jun 2020

Research & Skills

Research Interests: Uncertainty Quantification, Bayesian Optimization, Surrogate Modeling
Programming and Tools: R, Python, MySQL, SAS, MATLAB
Statistical Modeling: Regression, Mixed Models, Clustering, Classification, Gaussian Processes
Bayesian Methods: Variational Inference, Gibbs Sampling, MCMC, Bayesian Gaussian Mixture Models

Publication

Chang, C.-Y. and Sung, C.-L. (2025). Deep Intrinsic Coregionalization Multi-Output Gaussian Process Surrogate with Active Learning. *International Journal for Uncertainty Quantification*, accepted.

Experience

Research Assistant , Michigan State University East Lansing, MI	May 2021 - Present
• Designed a two-layer deep Gaussian process with mini-batch latent MAP estimation for scalable inference. • Developed a Multi-Output Deep Gaussian Process (MODGP) model with a coregionalization structure, achieving top-three predictive performance among nine benchmark emulators. • Designed active learning and sequential experimental designs for multi-output GP models, reducing predictive uncertainty and improving modeling efficiency.	
Teaching Assistant , Michigan State University East Lansing, MI	Sep. 2021 - Present
• Mentored over 500 students across introductory and advanced statistics courses, contributing to a 15% improvement in final exam performance through interactive and hands-on learning.	
Data Analyst Intern , Consilium Software Inc (PTE.) Ltd Taipei, Taiwan	Jul. 2019 - Aug. 2019
• Collaborated on developing a telemarketing analysis system in R, identifying key correlations between product attributes and sales outcomes. • Built a text-to-speech model, reducing processing time by 90% and improving customer needs analysis.	

Honors & Awards

NISS New Researcher Award (2026)
• Presented an invited talk on <i>Efficient Deep Gaussian Process Surrogates via Mini-Batch Latent MAP Estimation</i>
Dissertation Completion Fellowship (2026)
• Awarded by the College of Natural Science at Michigan State University in support of dissertation completion.
ASA/IMS Spring Research Conference Student Travel Award (2025)
• Delivered a research talk on <i>Deep Intrinsic Coregionalization Multi-Output Gaussian Process</i>.
IMSI Poster Acceptance (2025)
• Presented poster at the <i>Kernel Methods in Uncertainty Quantification and Experimental Design</i> program.
Harkness Outstanding Graduate Student Teaching Award (2023-2024)
• Selected from 30+ graduate TAs for exceptional contributions to statistics teaching.